A conformal-neck counterexample to scalar stability without connectivity

23 September 2026. This note proves a concrete limitation of the first-line functional-stability program. It first records a valid Ricci/connection bridge, then constructs metrics for which every energy on its right tends to zero while scalar variance and relative Einstein defect do not. All inputs are classical. No literature-priority claim is made.

The example stays on one fixed smooth manifold which already admits negative Einstein metrics. It is not a counterexample to Yau26.

1. The valid non-Kähler bridge, and the missing conclusion

Use the curvature conventions of Round 17 and the scalar-band note. Let an actual ordinary Seiberg–Witten solution have real determinant curvature $\mathscr{F}$, with $[\mathscr{F}/(2\pi)] = a$. Write

$$p = (a_g^+)^2 > 0, \quad T = 2\chi + 3\tau, \quad P = \int |\operatorname{Ric}^\circ|^2, \quad B_p = \int s^2 - 32\pi^2 p,$$
$$D = \frac{1}{4\pi^2}\int \left(\frac{s^2}{24} + \frac{|W^+|^2}{2}\right) - \frac{p}{2}.$$

Ordinary SW gives $B_p \geqslant 0$. Indeed, with $\rho = |\Phi|^2$, the exact Round-17 identity refines to

$$\boxed{B_p = \|s + \rho\|_2^2 + 8\|\nabla_A \Phi\|_2^2 + 8\left(\|\mathscr{F}^-\|_2^2 - 4\pi^2(p - a^2)\right).} \tag{1}$$

The last parenthesis is nonnegative by Hodge orthogonality. If $\mathscr{F}_H$ is the harmonic representative of $2\pi a$, exactness of $\mathscr{F} - \mathscr{F}_H$ equates the squared norms of its two Hodge components, so (1) also proves

$$\|\mathscr{F} - \mathscr{F}_H\|_2^2 \leqslant B_p/4, \qquad \|\nabla_A \Phi\|_2^2 \leqslant B_p/8. \tag{2}$$

These are controls of actual determinant curvature and an actual spinor, not a formal cohomology substitution. They do not assert control of the connection one-form modulo gauge without an additional elliptic estimate.

Chern–Weil gives the exact non-Kähler identity

$$\boxed{P + \frac{B_p}{4} = 8\pi^2\Big((p - a^2) + (a^2 - T)\Big) + 32\pi^2 D.} \tag{3}$$

Consequently small LB defect, small period excess, and small topological cost really do bound trace-free Ricci energy. If $a^2 \geqslant T$, all three terms on the right are nonnegative. This elementary classical consequence does not, however, bound scalar variance.

There is even a scalar-band estimate for B_p, independent of any Poincaré constant. Assume $-S \leqslant s \leqslant -s_0 < 0$, and the conformal-class SW solvability used in the scalar-band note. Put $w = \sqrt{6}|W^+|$ and $t = s + 2w$. Its retained defect split gives $\int t^2 \leqslant 288\pi^2 D$. Direct algebra gives

$$B_p = 64\pi^2 D + \frac{2}{3}\int st - \frac{1}{3}\int t^2,$$

$$\boxed{0 \leqslant B_p \leqslant 64\pi^2 D + 8\sqrt{2}\pi S\sqrt{VD}.} \tag{4}$$

Thus a fixed scalar band and $D/V \to 0$ also control the normalized energies in (1)–(2). The following example satisfies stronger hypotheses: $p = a^2 = T$ exactly, and its ordinary scalar defect tends to zero. Scalar variance nevertheless remains positive.

2. Fixed smooth topology and a separating hyperbolic collar

Fix integers $g_-, g_+ \geqslant 1$, $g = g_- + g_+$, and $h \geqslant 2$. Set

$$M = \Sigma_g \times \Sigma_h, \qquad A = 8(g-1)(h-1) = 2\chi(M) + 3\tau(M) > 0.$$

Choose a separating curve in Σ_g whose two sides have genera g_- and g_+. Hyperbolic pairs of pants, equivalently Fenchel–Nielsen coordinates, give curvature -1 metrics b_ℓ on this fixed marked surface for which this curve is a geodesic of length $\ell \downarrow 0$. Fix a curvature -1 metric h_F on Σ_h, and put

$$g_\ell = b_\ell \oplus h_F.$$

Then $\mathrm{Ric}_{g_\ell} = -g_\ell$, $s_{g_\ell} = -4$, and

$$V_0 = \mathrm{Vol}(g_\ell) = 16\pi^2(g-1)(h-1) = 2\pi^2 A. \tag{5}$$

The separating geodesic has an embedded collar

$$(-w_\ell, w_\ell) \times (\mathbb{R}/\mathbb{Z}), \qquad b_\ell = dt^2 + \ell^2 \cosh^2 t\, d\theta^2, \quad w_\ell = \operatorname{arsinh} \frac{1}{\sinh(\ell/2)} \to \infty. \tag{6}$$

This is the classical collar lemma; a primary research reference recalling it is Rupflin–Topping, Appendix A, Lemma A.1. Only (6), not a convergence theorem for moduli spaces, is needed below.

Cutting at $t = 0$ gives geodesic-boundary hyperbolic surfaces of areas $2\pi(2g_\pm - 1)$, by Gauss–Bonnet. The corresponding product-side volumes are therefore exactly

$$V_\pm = 8\pi^2(2g_\pm - 1)(h-1) > 0, \qquad V_- + V_+ = V_0. \tag{7}$$

3. A fixed-width transition and a uniform negative scalar band

Fix unequal positive constants c_-, c_+. Choose a smooth function $f : \mathbb{R} \to \mathbb{R}$ which equals $\log c_-$ for $t \leqslant -R$ and $\log c_+$ for $t \geqslant R$, and is constant on neighborhoods of those end intervals. Here $R < \infty$ is fixed. It can be chosen large enough that

$$6\Big(\|f''\|_\infty + \|f'\|_\infty + \|f'\|_\infty^2\Big) \leqslant 2. \tag{8}$$

Alternatively fix R first and choose $\log(c_+/c_-) \neq 0$ sufficiently small. For $w_\ell > R$, extend $f(t)$ to the two complementary sides by its respective constants; call the resulting smooth function f_ℓ on M, constant in the fiber direction. Set

$$\widehat{g}_\ell = e^{2f_\ell} g_\ell. \tag{9}$$

Use the Laplacian $\Delta = \operatorname{div}\nabla$, with nonpositive spectrum. On the transition region,

$$|df_\ell|^2 = (f')^2, \qquad \Delta f_\ell = f'' + \tanh(t) f', \qquad \hat{s}_\ell = e^{-2f_\ell}\left[-4 - 6\left(\Delta f_\ell + |df_\ell|^2\right)\right]. \tag{10}$$

Writing $f_{\min}, f_{\max}$ for the fixed extrema, (8) implies

$$-6e^{-2f_{\min}} \leqslant \hat{s}_\ell \leqslant -2e^{-2f_{\max}} < 0. \tag{11}$$

The support of the derivatives lies in $N_\ell = [-R, R] \times S^1 \times \Sigma_h$, whose unscaled volume is

$$\operatorname{Vol}_{g_\ell}(N_\ell) = 2\ell \sinh R \ \operatorname{Area}(h_F) = O(\ell). \tag{12}$$

All derivatives of f_ℓ of each fixed order are uniformly bounded: in the collar frame the coefficients are built from bounded derivatives of $\tanh t$ and the fixed function f. Hence the full curvature and each fixed covariant derivative of curvature of $\widehat{g}_\ell$ are uniformly bounded as well. Volumes are bounded above and away from zero. Injectivity radius and global connectivity are not uniformly controlled.

4. Actual monopole class, exact periods, and exact energy identities

The product almost-complex structures determined by the two oriented surface metrics are homotopic as ℓ varies. Choose the canonical or conjugate canonical Spin^c structure, and write $a = \pm c_1(K_M)$ for its real first Chern class. This is one fixed class and one fixed Spin^c structure. It has nonzero ordinary SW invariant; $b_+(M) = 1 + 2gh > 1$, so there is no chamber issue. Thus its unperturbed equations have solutions for all the metrics in (9), and for every metric conformal to each of them. The classical Kähler nonvanishing input is recorded in LeBrun, Theorem 2.

The Kähler form ω_ℓ of g_ℓ satisfies $[\omega_\ell] = 2\pi c_1(K_M)$. Hodge star on two-forms is unchanged under the conformal change (9); a closed self-dual form is harmonic for either metric. Consequently

$$\boxed{p_\ell = (a^+_{\widehat{g}_\ell})^2 = a^2 = A = T \quad \text{for every } \ell.} \tag{13}$$

Set

$$B_\ell = \int \hat{s}_\ell^2 \, d\widehat{\mu}_\ell - 32\pi^2 A, \qquad z_\ell = \Delta f_\ell + |df_\ell|^2.$$

The conformal factors cancel in the scalar-square integral. Since the manifold is closed, $\int \Delta f_\ell = 0$, and (10) gives the exact formula

$$\boxed{B_\ell = 48\int |df_\ell|^2\, d\mu_{g_\ell} + 36\int z_\ell^2\, d\mu_{g_\ell} > 0.} \tag{14}$$

The strict inequality holds because $c_- \neq c_+$. By (12), $B_\ell = O(\ell) \to 0$. This is the ordinary SW scalar defect from Round 17, not a new weighted defect.

The Weyl-square integral is conformally invariant in dimension four, and the starting Kähler–Einstein product has $\int |W^+|^2 = 2V_0/3$. Hence the LB defect satisfies the exact identity

$$\boxed{D_\ell = \frac{B_\ell}{96\pi^2} \longrightarrow 0.} \tag{15}$$

Subtract the Chern–Weil formula for g_ℓ from that for $\widehat{g}_\ell$, keeping this Weyl integral fixed. It gives

$$\boxed{P_\ell = \int |\,\mathring{\mathrm{Ric}}_{\widehat{g}_\ell}|^2 = \frac{B_\ell}{12} = 8\pi^2 D_\ell \longrightarrow 0,} \tag{16}$$

$$\boxed{R_{2,\ell} = 8\pi^2 A + \frac{B_\ell}{3} = 8\pi^2 A + 32\pi^2 D_\ell.} \tag{17}$$

These metrics also approach the absolute Ricci-energy infimum on this minimal general-type smooth manifold, by LeBrun, Theorem 4.3.

They are genuinely non-Kähler, not merely non-Kähler for the displayed complex structure. If $\widehat{g}_\ell$ were Kähler in the given orientation, $|W^+|^2 = \hat{s}_\ell^2/24$ would force $B_\ell = 0$, contradicting (14). The same argument applies in the other orientation: the starting product is Kähler for either orientation and has the same W^--energy. Thus $\widehat{g}_\ell$ is not Kähler for either orientation, although it is conformally Kähler for both.

5. The scalar variance and relative defect have positive limits

Define constants

$$H_0 = V_- + V_+, \qquad H_2 = V_- c_-^2 + V_+ c_+^2, \qquad H_4 = V_- c_-^4 + V_+ c_+^4. \tag{18}$$

Outside N_ℓ, the scalar is exactly $-4c_\pm^{-2}$ and the volume form is $c_\pm^4 d\mu_{g_\ell}$. The transition region has volume $O(\ell)$, uniformly bounded scalar, and bounded conformal factor. Using (7) gives

$$\widehat{V}_\ell \to H_4, \qquad \int \hat{s}_\ell\, d\widehat{\mu}_\ell \to -4H_2, \qquad \int \hat{s}_\ell^2\, d\widehat{\mu}_\ell \to 16H_0. \tag{19}$$

Therefore, with $L_\ell = \int (\hat{s}_\ell - \bar{\hat{s}}_\ell)^2$,

$$\boxed{L_\ell \longrightarrow 16\left(H_0 - \frac{H_2^2}{H_4}\right) = \frac{16V_-V_+(c_-^2 - c_+^2)^2}{H_4} > 0.} \tag{20}$$

Minimization over nonzero constant Einstein factors gives, as in Round 21,

$$\mathscr{E}(\widehat{g}_\ell) = \frac{4P_\ell + L_\ell}{R_{2,\ell}} \longrightarrow \boxed{4\left[1 - \frac{(V_-c_-^2 + V_+c_+^2)^2}{(V_-c_-^4 + V_+c_+^4)(V_- + V_+)}\right]} > 0. \tag{21}$$

For instance, take $g_- = g_+ = 1$, $h = 2$, $c_- = 1$, and $c_+ = \sqrt{2}$. The two side volumes are equal and the limiting relative defect is exactly $2/5$. A sufficiently wide but fixed transition gives the negative scalar band (11).

The example can also be globally rescaled to mean scalar -4. Indeed $\kappa_\ell = -\bar{\hat{s}}_\ell/4 \to H_2/H_4 > 0$, and the metric $\kappa_\ell \widehat{g}_\ell$ has that mean scalar. All energies in (13)–(21) are unchanged by this constant rescaling; the scalar band remains uniform. With the mean normalization, the averaged Einstein residual obeys

$$\frac{\int |\mathrm{Ric} + \kappa_\ell \widehat{g}_\ell|^2}{\kappa_\ell^2 \widehat{V}_\ell} \longrightarrow 4\left(\frac{H_0 H_4}{H_2^2} - 1\right) > 0. \tag{22}$$

Thus the failure is not removed by fixing the mean scalar or the volume scale. The missing information concerns communication between the two bulk regions.

6. The first eigenvalue and the actual SW amplitude

Use f_ℓ, after subtracting its $\widehat{g}_\ell$-mean, as a scalar Rayleigh test. Its Dirichlet energy is

$$\int |df_\ell|^2_{\widehat{g}_\ell} d\widehat{\mu}_\ell = \int e^{2f_\ell} |df_\ell|^2_{g_\ell} d\mu_{g_\ell} = O(\ell),$$

whereas its variance has a positive limit because its two bulk constants are unequal. Hence

$$\lambda_1(\widehat{g}_\ell) = O(\ell) \longrightarrow 0. \tag{23}$$

There is no conflict with actual SW control. Since $p = a^2$, the right side of the ordinary identity (1) is precisely $B_\ell \to 0$. Any ordinary solution satisfies $\|\hat{s}_\ell + |\Phi_\ell|^2\|_2 \to 0$ and $\|\nabla_{A_\ell}\Phi_\ell\|_2 \to 0$, while its amplitude follows the two different bulk scalar values. Small gradient energy does not force a single global amplitude across this degenerating bottleneck.

Likewise, the harmonic representative of the canonical class is the same differential form $\pm\omega_\ell/(2\pi)$ before and after (9). Its pointwise norm after conformal scaling is $\sqrt{2}e^{-2f_\ell}/(2\pi)$, with two different bulk limits, even though its squared L^2 norm remains exactly $p = A$. Primitive-class control fixes a cohomological norm; it does not fix the pointwise amplitude.

7. Optional stronger variant: pointwise trace-free Ricci tends to zero

The fixed-window proof already suffices. One may also let $R_\ell = \frac{1}{2}\log(1/\ell) \to \infty$, and choose

$$f_\ell(t) = \log c_- + \log(c_+/c_-)\,\chi(t/R_\ell),$$

where χ is a fixed smooth step equal to zero for inputs at most -1 and one for inputs at least 1. For small ℓ, this fits inside (6). Its transition volume is $O(\ell e^{R_\ell}) = O(\sqrt{\ell})$, and

$$\|df_\ell\|_\infty = O(R_\ell^{-1}), \qquad \|\nabla^2 f_\ell\|_\infty = O(R_\ell^{-1}).$$

For the Hessian, the collar eigenvalues are f'', $\tanh(t)f'$, and two zeros. The conformal trace-free Ricci formula is

$$\mathrm{Ric}^{\circ}_{\widehat{g}_\ell} = -2\Big(\nabla^2 f_\ell - df_\ell \otimes df_\ell\Big)_0. \tag{24}$$

The trace-free operation may be taken with g_ℓ, since it is unchanged under conformal scaling of the metric. The conformal factors are uniformly bounded above and below, so $\|\mathrm{Ric}^{\circ}_{\widehat{g}_\ell}\|_\infty = O(R_\ell^{-1}) \to 0$. The scalar band holds for all sufficiently small ℓ. Formula (14) gives $B_\ell = O(\sqrt{\ell}/R_\ell^2) \to 0$, and all limits (19)–(22) remain unchanged. Thus even pointwise convergence of the trace-free Ricci tensor to zero does not produce one global Einstein factor here.

8. Precise consequence and non-consequence

This disproves a uniform estimate tending to zero for scalar variance or relative Einstein defect based only on LB defect, ordinary SW scalar defect, period excess, topological cost, a fixed negative scalar band, bounded volume, and even uniform local curvature bounds. All those inputs have the required uniform bounds or vanish along this fixed-topology sequence. A scalar spectral gap or another genuinely global connectivity hypothesis cannot simply be deleted from the scalar-band stability theorem.

It does not disprove the valid Ricci/connection bridge (1)–(4), the classical exact equality theorem, or stability under stronger compactness conditions. Nor is it a Yau26 counterexample: each M here admits the original product Einstein metric. Finally, the construction is an explicit application of classical conformal and hyperbolic geometry; no assertion is made that this mechanism or its mathematical conclusion is new in the literature.